Grade 9 Computer Science

Solved Exercises - English Medium - Balochistan Textbook Board (BTBB)

Complete book: all 9 units, all exercises (multiple choice, short answers and detailed answers).

Unit 1: Computer Systems

Unit 2: Computer Software

Unit 3: Data Communication and Networks

Unit 4: Computational Thinking and Algorithms

Unit 5: Web Designing

Unit 6: Data and Analysis

Unit 7: Applications of Computer Science

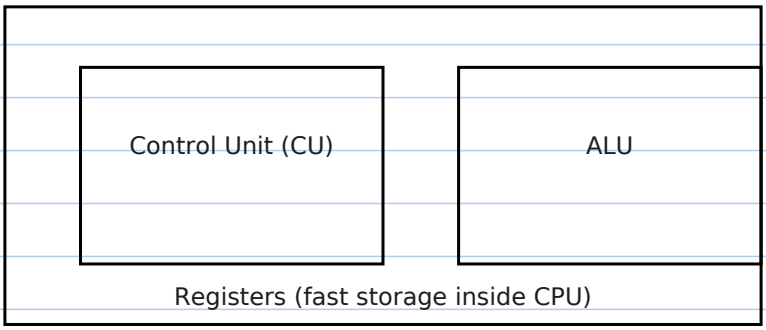
Unit 8: Impacts of Computing

Unit 9: Entrepreneurship in Digital Age

Source: BTBB Grade 9 Computer Science textbook (ASAN, Single National Curriculum 2022-23), 220 pages. Medium confirmed from the book itself: the cover, contents and all lessons are in English. Questions are reproduced from each unit exercise; answers are written in medium length. Where a book question contains an error (Unit 3 asks for 'five' OSI layers while the book teaches seven), the answer follows the book and notes the difference.

Unit 1: Computer Systems

Diagram: The CPU and its parts



Exercise 1

1. Multiple choice questions. (Correct option marked with a green tick.)

(i) Which generation of computers came into existence with the invention of transistors?

(a) First

(b) Second ✓

(c) Third

(d) Fourth

(ii) The fourth generation of computers is based on:

(a) ICs

(b) Transistors

(c) Vacuum Tubes

(d) Microprocessor ✓

(iii) Which is the characteristic of a natural system?

(a) Existence in nature ✓

(b) Controlled by human

(c) Controlled by machines

(d) Purposeful design

(iv) Which is the characteristic of an artificial system?

(a) Existence in nature

(b) Controlled by human ✓

(c) Controlled by machines

(d) Work together as a unit

(v) Which parts of a computer are considered hardware?

(a) Technical

(b) Physical ✓

(c) Non-Physical

(d) Natural

(vi) Software is the set of:

(a) Data

(b) Processing

(c) a and b

(d) Instructions ✓

(vii) A computer stores the data in:

(a) Keyboard

(b) Printer

(c) Monitor

(d) Memory ✓

(viii) Which of the following is not placed in the System Unit?

(a) RAM

(b) Sound Card

(c) Scanner ✓

(d) Hard Disk

(ix) The brain of the computer system:

(a) Information

(b) CPU ✓

(c) Monitor

(d) None of these

(x) All hardware devices are attached directly or indirectly with:

(a) Monitor

(b) Printer

(c) Motherboard ✓

(d) Modem

2. Short answers.

(i) How do input and output devices differ? Provide two examples of each.

Input devices send data INTO the computer, while output devices bring processed results OUT of it. Input examples: keyboard and mouse (also scanner). Output examples: monitor and printer (also speakers). Input devices are used to give instructions and data; output devices present information to the user.

(ii) What is the name of the device used to store data on a computer?

Data is stored in memory and storage devices. Primary memory (RAM) holds data temporarily while work is in progress, and secondary storage devices such as a hard disk, SSD or USB drive store data permanently.

(iii) What are the four main components that make up a computer system?

The four main components are input devices, the processing unit (CPU), memory/storage, and output devices. Together they take in data, process it, store it and present the result.

(iv) In which units is the storage capacity of a hard drive measured?

Hard drive capacity is measured in bytes and its multiples: kilobytes (KB), megabytes (MB), gigabytes (GB) and terabytes (TB). Modern hard drives are usually rated in GB or TB.

(v) What is the most common keyboard layout?

The most common keyboard layout is QWERTY, named after the first six letters on the top row of letter keys. It is the standard layout on most computers.

(vi) Write a short note on optical devices.

Optical devices use light and lenses to read or show information. Optical storage devices such as CDs, DVDs and Blu-ray discs store data as tiny marks read by a laser beam. Optical input devices include scanners and webcams, which capture images using light.

(vii) How many expansion slots are there in a computer? Write their names.

The number of expansion slots depends on the motherboard, but the common types are PCI, PCI Express (PCIe), AGP (older graphics slot) and ISA (very old). Modern computers mainly use PCIe slots for graphics, sound and network cards.

(viii) Write the name of major expansion cards.

Major expansion cards are the graphics (video) card, sound card, network interface card (NIC) and modem card. Each adds a function that the motherboard does not provide on its own.

(ix) What do you know about cache memory?

Cache memory is a small, very fast memory built into or near the CPU. It stores copies of data and instructions the CPU uses most often, so the CPU does not have to wait for slower main memory. This speeds up processing.

(x) How does the processor perform its operation?

The processor works in a fetch-decode-execute cycle. It fetches an instruction from memory, decodes what it means, executes it in the ALU or control unit, and stores the result. This cycle repeats millions of times per second.

3. Detailed answers.

(i) What are the fundamental components of a computer system? Explain.

A computer system has hardware and software working together. Hardware includes input devices (keyboard, mouse), the system unit (motherboard, CPU, memory, power supply), storage devices (hard disk, SSD) and output devices (monitor, printer). Software includes the operating system and application programs that tell the hardware what to do. Data flows in through input, is processed by the CPU, stored in memory, and shown through output.

(ii) Distinguish between the hardware and software layers of a computer in detail.

Hardware is the physical part of a computer that can be touched, such as the CPU, memory, keyboard and monitor. Software is the set of programs and instructions that cannot be

touched but controls the hardware, for example Windows and MS Word. Hardware alone cannot work without software, and software cannot run without hardware; the operating system forms the layer between user programs and hardware.

(iii) Explain how the CPU functions within a computer system.

The CPU is the brain of the computer. Its Control Unit (CU) fetches instructions and directs all parts of the computer. Its Arithmetic Logic Unit (ALU) performs calculations and logical comparisons. Registers are tiny, fast storage locations inside the CPU that hold data being processed. Together they run the fetch-decode-execute cycle that carries out every program.

(iv) Describe how different computer components work together to process and transfer data and instructions.

Data enters through an input device and is held in main memory. The CPU fetches instructions from memory, processes the data, and sends the result back to memory or storage. Buses carry data and instructions between the CPU, memory and devices, and the output device finally presents the result to the user. The operating system coordinates all these steps.

Unit 2: Computer Software

Exercise 2

1. Multiple choice questions. (Correct option marked with a green tick.)

(i) The set of instructions given to the computer to perform a specific task is called:

(a) Hardware

(b) Software ✓

(c) Anti-Virus

(d) E-Mail

(ii) Which of the following creates a link between a user and the computer?

(a) Device Driver

(b) Output Devices

(c) Operating System ✓

(d) Utility

(iii) Which of the following is the most popular type of operating system for a personal computer?

(a) DOS

(b) Linux

(c) Unix

(d) MS Windows ✓

(iv) A system software that helps the operating system to communicate with a device is called:

(a) Operating System

(b) Language Translator

(c) Device Driver ✓

(d) Utility

(v) Which of the following is the program that allows a user to analyse and maintain a computer?

(a) Device Driver

(b) Utility ✓

(c) Windows XP

(d) File Viewer

(vi) A computer software that helps users to do a specific task on a computer is called:

(a) Application Software ✓

(b) Utility

(c) System Software

(d) MS Office

(vii) Computer games are an example of:

(a) Entertainment Software ✓

(b) Reference Software

(c) System Software

(d) Multimedia Software

(viii) Which of the following allows us to communicate with the computer?

(a) Business Software

(b) Programming Languages ✓

(c) Productivity Software

(d) Games

(ix) Which of the following software are used to create visual presentations?

(a) MS PowerPoint ✓

(b) MS Paint

(c) Typing Tutor

(d) MS Word

(x) Researching to develop more user-friendly programs is the responsibility of:

(a) Hardware Engineer

(b) End User

(c) Software Engineer ✓

(d) Computer Operator

2. Short answers.

(i) Define software.

Software is a set of instructions, programs and data that tells a computer what to do and how to do it. Without software, computer hardware cannot perform any useful task.

(ii) Differentiate between application software and system software.

System software manages the computer itself, for example the operating system, device drivers and utilities. Application software helps the user do a specific job, for example word processing, spreadsheets, browsers and games. System software runs the machine; application software serves the user.

(iii) What is an operating system?

An operating system is the main system software that manages all hardware and software resources of a computer. It acts as a link between the user and the hardware, and controls memory, processes, files and devices.

(iv) Differentiate between word processing and spreadsheet.

Word processing software (for example MS Word) is used to create, edit and format text documents such as letters and reports. Spreadsheet software (for example MS Excel)

organises data in rows and columns and performs calculations with formulas.

(v) List the names of any four application software.

Four application software are: MS Word (word processing), MS Excel (spreadsheet), MS PowerPoint (presentation) and Google Chrome (web browsing). VLC media player and computer games are further examples.

(vi) List the names of language translators.

The main language translators are the assembler, the compiler and the interpreter. An assembler translates assembly language, a compiler translates a whole high-level program at once, and an interpreter translates and runs it line by line.

(vii) Can a device work without a device driver?

No, a device normally cannot work properly without its device driver. The driver tells the operating system how to communicate with that particular device; without it the operating system cannot control the device correctly.

3. Detailed answers.

(i) What is computer software? Discuss their types in detail.

Computer software is the set of programs and instructions that control a computer. Its two main types are system software and application software. System software includes the operating system, device drivers, language translators and utilities, and it manages the hardware. Application software includes word processors, spreadsheets, presentation software, browsers, media players and games, and it helps the user complete specific tasks.

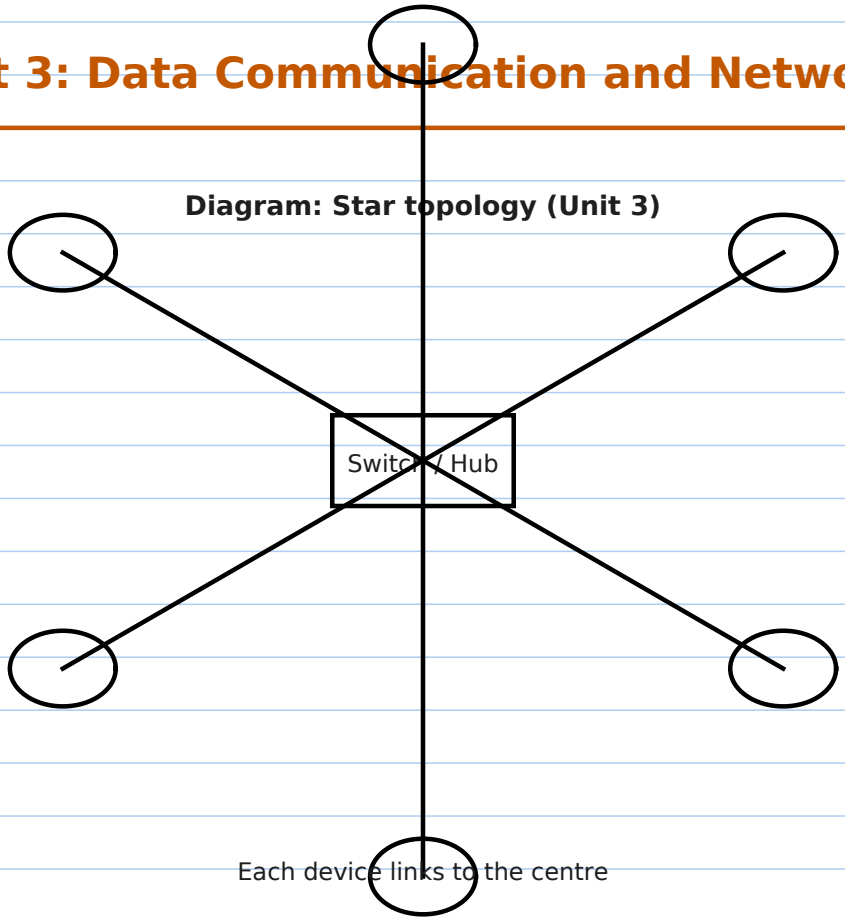
(ii) Describe the main operations of an operating system.

An operating system performs process management (running and scheduling programs), memory management (giving programs the memory they need), file management (storing and organising files), device management (controlling hardware through drivers) and user-interface management (providing the screen, menus and commands the user works with).

(iii) What are the different uses of various types of application software?

Word processors are used to write documents, spreadsheets to calculate and analyse numbers, presentation software to make slides, browsers to use the internet, media players to play audio and video, and entertainment software such as games for leisure. Business, education and reference software serve offices, schools and research in the same way.

Diagram: Star topology (Unit 3)



Exercise 3

1. Multiple choice questions. (Correct option marked with a green tick.)

(i) Which of the following is a communication device?

(a) Mouse

(b) Hard disk

(c) Printer

(d) Modem ✓

(ii) Which of the following network is confined to a relatively small area?

(a) WAN

(b) MAN

(c) Internet

(d) LAN ✓

(iii) Which of the following device is used to send messages, data or information to other devices?

(a) Receiving device

(b) Sending device ✓

(c) Input device

(d) Communication device

(iv) VPN stands for:

(a) Vital Private Network

(b) Virtual Package Network

(c) Virtual Private Network ✓

(d) Vocal Private Network

(v) LAN stands for:

(a) Large Area Network

(b) Local Area Network ✓

(c) Local Arena Network

(d) Latest Area Network

(vi) Which mode of network allows devices to send and receive data at the same time?

(a) Simplex mode

(b) Half duplex mode

(c) Full duplex mode ✓

(d) All of these

(vii) Which of the following topologies uses point-to-point connections?

(a) Ring ✓

(b) Star

(c) Bus

(d) All of these

(viii) Which protocol is used for transferring web pages on the internet?

(a) FTP

(b) IP

(c) HTTP ✓

(d) SMTP

2. Short answers.

(i) What is a computer network, and how does it differ from a standalone computer?

A computer network is a group of two or more computers and devices connected so they can share data and resources. A standalone computer works alone and cannot share files, printers or internet access with other computers.

(ii) How is the Internet an example of a computer network, and how does it work?

The Internet is the largest computer network in the world, joining millions of smaller networks together. Computers send data in packets through routers and servers using standard rules (protocols such as TCP/IP), and the packets are reassembled at the destination.

(iii) What are LAN and WAN, and what is the difference between them?

A LAN (Local Area Network) covers a small area such as one room, school or office and is fast and privately owned. A WAN (Wide Area Network) covers a large area such as cities or countries, is slower, and is usually run by several organisations. The Internet is the largest WAN.

(iv) What are the different types of transmission media used in networking?

Transmission media are guided (wired) media such as twisted-pair cable, coaxial cable and fibre-optic cable, and unguided (wireless) media such as radio waves, microwaves, infrared and satellite links.

(v) What are the layers of the OSI reference model, and what is the function of each layer? (Note: the book's OSI model has seven layers, not five.)

The seven OSI layers are: Physical (sends bits over the medium), Data Link (reliable link, MAC addresses, error detection), Network (best path and IP addressing, where routers work), Transport (end-to-end delivery), Session (manages a session between applications), Presentation (data format and encryption) and Application (services for the user, such as web and email).

(vi) In which layer of the OSI model does a router operate?

A router operates at the Network layer (Layer 3) of the OSI model, because it reads IP addresses and chooses the best path for packets between networks.

(vii) How does circuit switching differ from packet switching?

In circuit switching a complete, dedicated path is reserved for the whole communication, as in a traditional telephone call, so there is no delay once it is set up but the path stays busy. In packet switching data is broken into small packets that travel independently, possibly by different routes, and are reassembled at the destination, as on the Internet.

(viii) What is the purpose of using bridges in a computer network?

A bridge connects two similar network segments and forwards only the traffic meant for the other segment. This reduces unnecessary traffic, divides a large network into smaller parts and improves performance.

(ix) Differentiate between DNS and DHCP.

DNS (Domain Name System) translates easy domain names such as www.example.com into IP addresses that computers use. DHCP (Dynamic Host Configuration Protocol) automatically gives computers on a network an IP address and related settings, so they do not have to be set by hand.

(x) How does the Internet work?

The Internet works by breaking messages into packets that travel through routers, switches and servers using TCP/IP rules. Each packet carries the destination address, different packets may take different routes, and the receiving computer reassembles them into the original message.

3. Detailed answers.

(i) Explain different types of computer networks.

Networks are classified by size: PAN (personal area, a few metres around one person), LAN (a room, building or campus), MAN (a city) and WAN (cities, countries or the world). The Internet is the largest WAN. Larger networks cover more area but are usually slower and

more costly than a LAN.

(ii) Discuss different components of a computer network.

A network needs devices (computers, printers, phones), network devices (router, switch, modem, hub, bridge), transmission media (cables or wireless links), and software/protocols (such as TCP/IP) that set the rules of communication. Servers store and share resources, while clients are the computers that use them.

(iii) Define network communication. Explain the different modes of network communication with examples.

Network communication is the exchange of data between connected devices. In simplex mode data flows in one direction only, for example a keyboard or a radio broadcast. In half-duplex mode data flows in both directions but not at the same time, for example a walkie-talkie. In full-duplex mode data flows in both directions at the same time, for example a telephone call or video call.

(iv) Describe network topology. How does ring topology differ from star topology?

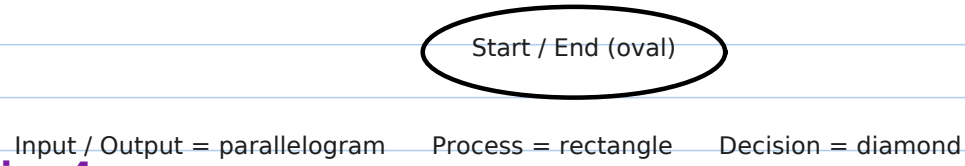
Topology is the arrangement of devices and links in a network. In a ring topology devices are connected in a closed loop and data travels around the ring, so one device failure can affect the whole network. In a star topology every device connects to a central switch or hub, so one device failure does not affect the others, but failure of the central device stops the whole network.

(v) Explain the OSI layers model with the help of a real-world example.

The OSI model is like sending a letter through a postal system, where each layer is one office with one job: the Application layer is the person writing the letter, Presentation translates and seals it, Session opens the conversation, Transport checks delivery, Network chooses the route, Data Link delivers it to the next post office, and Physical carries it on the road. Each layer passes the message to the next until it reaches the receiver, where the steps run in reverse.

Unit 4: Computational Thinking and Algorithms

Diagram: Flowchart symbols (Unit 4)



Exercise 4

1. Multiple choice questions. (Correct option marked with a green tick.)

(i) Which is a process of identifying a problem and finding its best solution?

(a) Decomposition

(b) Ideal solution

(c) Problem Solving ✓

(d) Challenge

(ii) In which phase do we need to understand and clearly define the problem?

(a) Problem definition

(b) Potential solution

(c) Problem identification ✓

(d) Condition statement

(iii) Decomposition reduces a problem into:

(a) sub-problems ✓

(b) Complex problems

(c) Constraints

(d) Potential solution

(iv) An ideal solution among the potential solutions is to be selected as:

(a) First option

(b) Second option

(c) Third option

(d) Last option ✓

(v) Identify the steps of the problem-solving process.

(a) Four

(b) Six ✓

(c) Eight

(d) Nine

(vi) Which is the process that turns the ideal solution into actions?

(a) Decomposition

(b) Evaluation

(c) Implementation ✓

(d) Analyzing

(vii) Which is the sequence of steps to solve the problem in a precise and systematic way?

(a) Program

(b) Algorithm ✓

(c) Analysis

(d) Output

(viii) Which of the following refers to the act of doing one thing again and again?

(a) Repetition ✓

(b) Algorithm

(c) Solution

(d) Problem

(ix) Which is not a property of an algorithm?

(a) Output

(b) Generality

(c) Infiniteness ✓

(d) Effectiveness

(x) Why do we need to think computationally?

(a) To help us program

(b) To solve complex problems easily ✓

(c) To help us think like a computer

(d) To help us think like a genius

2. Short answers.

(i) Define problem solving.

Problem solving is the process of identifying a problem and finding its best solution. It means understanding the problem clearly, thinking of possible ways to solve it, choosing the best way, and checking whether it worked.

(ii) How can we identify a computing problem?

We identify a computing problem by asking: what is the problem, can a computer solve it, who is facing it, and what result is needed? If the task can be described by inputs, a process and an output, it is a computing problem.

(iii) Name the techniques of decomposition.

The techniques of decomposition are abstraction, mind mapping, divide and conquer, refinement and modularisation. Each technique breaks a big problem into smaller, easier parts in a different way.

(iv) Define computational thinking.

Computational thinking is the process of identifying a clear, defined, step-by-step solution to a complex problem, thinking like a computer scientist. It uses decomposition, pattern

recognition, abstraction and algorithmic thinking.

(v) Write an example of an algorithm from a real-world problem.

Algorithm to change a flat tyre: Step 1: Park safely and apply the handbrake. Step 2: Take out the jack and spare tyre. Step 3: Loosen the wheel nuts. Step 4: Jack up the car. Step 5: Remove the flat tyre. Step 6: Fit the spare tyre and tighten the nuts. Step 7: Lower the car and check the nuts.

(vi) What is the purpose of an algorithm?

The purpose of an algorithm is to give a clear, step-by-step plan for solving a problem before coding begins. It removes confusion, reduces errors, makes the solution easy to understand and to repeat, and helps the programmer write code faster.

(vii) How does the process of implementation differ from the process of evaluation in problem-solving?

Implementation is putting the chosen solution into action, for example actually fixing the printer. Evaluation is checking afterwards whether the solution worked, by testing it and seeking feedback, and trying the next solution if it did not.

(viii) Differentiate between pattern recognition and abstraction.

Pattern recognition means looking for things that repeat or are similar in different situations, which helps predict what will happen next. Abstraction means focusing only on the important details and ignoring unimportant ones, so the main idea of the problem stays clear.

(ix) Define flowchart with a real-world problem.

A flowchart is a diagram that shows the steps of a process using standard symbols connected by arrows. For a train gate: Start, then the sensor checks 'Train arriving?' If yes, close the gate; then check 'Train departed?' If yes, open the gate; then Stop.

(x) Draw a flowchart to find the sum, product and average of five numbers.

Flowchart: Start (oval) -> Input five numbers a, b, c, d, e (parallelogram) -> sum = a+b+c+d+e; product = a*b*c*d*e; average = sum/5 (rectangles) -> Print sum, product, average (parallelogram) -> Stop (oval). Arrows run top to bottom, one flow line entering each decision and start/end symbol.

3. Detailed answers.

(i) What are problem-solving methods, and how can they be defined and described?

The six-step problem-solving process is: (1) Identify and define the problem, (2) determine the root cause, (3) develop potential solutions, (4) select the ideal solution, (5) implement the solution, and (6) evaluate the outcome. Decomposition, abstraction and pattern recognition support these steps by making a complex problem smaller and clearer before a solution is chosen.

(ii) What is the role of computational thinking in computer science?

Computational thinking trains the mind to solve problems logically and systematically, the way a computer scientist does. It helps students break problems down, spot patterns, ignore irrelevant detail and design repeatable step-by-step solutions that can be turned into programs.

(iii) How can computational thinking principles be applied to define and refine problems?

First decompose the problem into small parts, then look for patterns among those parts, then abstract away unimportant details to state the problem precisely, and finally write the solution as an algorithm. Testing the algorithm on examples refines the problem statement until the input, process and output are all clear.

(iv) How can we evaluate whether the order of activities will lead to the desired outcome?

We evaluate the order by tracing the steps in sequence with sample inputs and checking each intermediate result against the expected result. If every step produces the right result and the final output matches the goal, the order is correct; otherwise the steps must be reordered or corrected and tested again.

(v) How can we write logical solution steps, and create a concept map and flowchart for solving a word search puzzle, calculator program, or tic-tac-toe game?

For a calculator program: Step 1: Start. Step 2: Input two numbers and an operator. Step 3: If the operator is '+', add; if '-', subtract; if '*', multiply; if '/', divide. Step 4: Display the result. Step 5: Stop. A flowchart uses an oval for Start/Stop, a parallelogram for input/output, rectangles for calculations and a diamond for the operator decision. A concept map links the main idea (Calculator) to concepts (Input, Operator, Calculation, Output) with labelled arrows showing how they relate.

Unit 5: Web Designing

Diagram: Layer cake of web technologies (Unit 5)



Exercise 5

1. Multiple choice questions. (Correct option marked with a green tick.)

(i) Every webpage has a unique _____.

(a) color

(b) logo

(c) URL ✓

(d) code

(ii) How many types of websites are there?

(a) Two ✓

(b) Three

(c) Five

(d) Six

(iii) Which of the following is a computer program that utilizes web browsers and web technology to perform tasks over the Internet?

(a) Program

(b) Web Application ✓

(c) Web Page

(d) Web Link

(iv) Which of the following is made up of a fixed number of pre-built files stored on a web server?

(a) First Page

(b) Dynamic Website

(c) Static Website ✓

(d) None of these

(v) Identify the server-side portion of website development.

(a) Front End Development

(b) Back End Development ✓

(c) Full Stack Development

(d) Client Server Model

(vi) What is the point at which the website is published?

(a) Testing

(b) Maintenance

(c) Implementation

(d) Deployment ✓

(vii) CSS stands for:

(a) Complete Style Sheet

(b) Complex Style Sheet

(c) Cascade Style Sheet ✓

(d) Coding Style Sheet

(viii) Visual Studio is an example of _____.

(a) Website

(b) Flowchart

(c) Game

(d) IDE ✓

(ix) How many heading tags are available in HTML?

(a) Two

(b) Four

(c) Six ✓

(d) Eight

(x) Which is the third layer of the layer cake of standard web technologies?

(a) HTML

(b) CSS

(c) C++

(d) JavaScript ✓

2. Short answers.

(i) Differences between a website and a web application.

A website is mainly a collection of webpages that present information, such as a school site. A web application is a program that runs in a browser and lets the user perform tasks, such as Gmail or an online shop, with data changing according to what the user does.

(ii) Difference between a static and a dynamic website.

A static website is made of fixed, pre-built files that look the same for every visitor and change only when edited by hand. A dynamic website builds its pages when requested, so content can change for each user, for example showing a logged-in user's name or latest news.

(iii) What is the role of an IDE in web development?

An IDE (Integrated Development Environment), such as Visual Studio Code, gives the developer one place to write code, with colour highlighting, error checking, file management and a preview or run option. This makes writing and debugging HTML, CSS

and JavaScript faster and easier.

(iv) What does HTML mean and what is it used for?

HTML stands for HyperText Markup Language. It is used to create the structure of a webpage, using tags to mark headings, paragraphs, links, images and lists, which the browser then displays.

(v) How can you open a link in a new window with HTML?

Use the target attribute with the value _blank in the anchor tag, for example: Visit. Clicking the link then opens the page in a new browser tab or window.

(vi) Which tag is used to create an ordered list?

The <ol> tag creates an ordered (numbered) list, and each item inside it is written with an <li> tag. For example: First
Second shows items numbered 1, 2.

(vii) What is CSS and what is it used for?

CSS stands for Cascading Style Sheets. It is used to style a webpage: colours, fonts, sizes, spacing and layout. HTML gives the structure, while CSS controls how that structure looks.

(viii) What is the correct CSS syntax for making all the elements bold?

span { font-weight: bold; } — the selector span picks every element, and the declaration font-weight: bold makes its text bold.

(ix) What are the different types of loops? Describe with an example.

The main JavaScript loops are: for loop (repeats a known number of times, for example for (let i = 1; i <= 5; i++)), while loop (repeats while a condition is true), and do-while loop (runs once, then repeats while the condition is true). A for...of loop can also run through the items of a list.

(x) How is debugging performed using the console.log() method?

Debugging with console.log() means placing console.log(variable) at points in the code to print values to the browser console. By reading those printed values the developer can see where a value goes wrong and find the faulty step.

3. Detailed answers.

(i) What is the difference between static and dynamic websites? Illustrate with an example.

A static website shows the same fixed pages to every visitor, for example a small company brochure site whose pages only change when the owner edits the files. A dynamic website creates pages on demand from a database, for example a news site or online shop that shows different headlines, prices or user accounts to different visitors at different times.

(ii) What is a web browser? Explain what happens when you type a website address in a browser.

A web browser (such as Chrome or Firefox) is software that finds, displays and lets you

interact with webpages. When you type an address, the browser asks DNS for the site's IP address, sends an HTTP request to that server, receives the HTML, CSS and JavaScript files, and renders them as the visible page.

(iii) What are the main benefits of using JavaScript?

JavaScript makes webpages interactive without reloading the page: it can check forms, respond to clicks, show or hide content, update parts of a page, and run games and web applications. It runs in every modern browser, so no extra software is needed.

(iv) Inside which HTML element do we put the JavaScript? If yes, then how?

JavaScript is put inside the <script> element. It can go in the <head> or at the end of the <body>, for example: <script> alert("Hello"); </script>. Placing it at the end of the body is common so the page content loads first.

(v) What is the correct syntax for referring to an external script called "xxx.js"?

<script src="xxx.js"></script> — the src attribute names the external file, and the browser loads and runs that file when the page loads.

(vi) What are some basic data types in JavaScript? Give an example usage of each.

Basic JavaScript data types are: Number (let age = 15;), String (let name = "Ali";), Boolean (let passed = true;), Undefined (a declared variable with no value), Null (an intentional empty value), and Object/Array (let marks = [80, 90, 70];). Each type suits a different kind of value.

(vii) Which of the tags below must be located in the <head> section of your page? <title> <form> <link> <meta>

The tags that belong in the <head> section are <title>, <link> and <meta>. <title> gives the page title shown on the browser tab, <link> connects an external stylesheet, and <meta> gives page information such as character set. <form> does NOT belong in the head; it goes in the <body> where the user fills it in.

Unit 6: Data and Analysis

Exercise 6

1. Multiple choice questions. (Correct option marked with a green tick.)

(i) Which are the observations or measurements (processed or unprocessed) expressed as text, numbers, or multimedia?

(a) Data ✓

(b) Database

(c) Dataset

(d) Data Science

(ii) Which is a collection of related data, generally in the form of a table or spreadsheet, that is primarily used for analysis?

(a) Data

(b) Database

(c) Dataset ✓

(d) Data Science

(iii) Which of the following is a discipline that combines multiple subject areas to process big data and determine patterns and trends?

(a) Computer Science

(b) Statistics

(c) Mathematics

(d) Data Science ✓

(iv) Which of the following refers to any machine that exhibits human-like characteristics such as learning and problem-solving?

(a) Computer Science

(b) Artificial Intelligence ✓

(c) Machine Learning

(d) Data Science

(v) Which of the following is a subset of machine learning that relies on artificial neural network design?

(a) Robotics

(b) Artificial Intelligence

(c) Deep Learning ✓

(d) Data Mining

(vi) Which relies on effective data collection, warehousing, and computer processing?

(a) Data Mining ✓

(b) Machine Learning

(c) Data Set

(d) Deep Learning

(vii) Data are mainly classified into ____ types.

(a) Two ✓

(b) Four

(c) Five

(d) Ten

(viii) Which is the numerical data that represents an unspecified number of measurements between two realistic points?

(a) Discrete

(b) Continuous ✓

(c) Ordinal

(d) Nominal

(ix) Which data collection is expensive and time consuming?

(a) Primary ✓

(b) Secondary

(c) Both a & b

(d) None of these

(x) Which includes gathering and interpreting non-numerical facts?

(a) Quantitative Data

(b) Qualitative Data ✓

(c) Primary Data

(d) Secondary Data

2. Short answers.

(i) Identify the key concepts and principles of data science.

Data science combines collecting data, cleaning it, analysing it for patterns, and presenting the results so decisions can be made. Its key principles are: use reliable data, ask a clear question, analyse honestly without bias, protect privacy, and present results clearly.

(ii) Define data. Give an example of something that is not data.

Data are observations or measurements, processed or unprocessed, expressed as text, numbers or multimedia, for example a student's marks or a photograph. A personal opinion stated without any observation or measurement, such as 'this is the best book', is not data by itself.

(iii) Define a database, its rationale, and its purpose for managing data.

A database is an organised collection of related data stored so it can be found, updated and used easily. Its purpose is to avoid repeated and inconsistent copies, keep data safe, allow many users to share it, and make searching and reporting fast and reliable.

(iv) Define and differentiate between data science and computer science.

Computer science is the study of computers, algorithms, programming and systems. Data science uses computer science together with mathematics, statistics and subject knowledge to process large data and find patterns and trends that support decisions. Computer science builds the tools; data science uses them on data.

(v) Differentiate between qualitative and quantitative data.

Qualitative data describe qualities and are non-numerical, for example colours, opinions or interview answers. Quantitative data are numerical and can be counted or measured, for example height, marks or number of students, and can be analysed with calculations.

(vi) List some sources for secondary data collection.

Sources of secondary data (data collected by others) include government reports and census data, books and newspapers, previous research studies, company records, and websites and databases published by organisations.

(vii) Differentiate between nominal and ordinal data.

Nominal data name categories with no natural order, for example gender or blood group. Ordinal data also name categories, but the categories have a meaningful order, for example first, second and third position, or ratings such as poor, good and excellent.

(viii) What is the use of deep learning in data analytics?

Deep learning, which uses artificial neural networks with many layers, finds complex patterns in very large data, such as recognising faces in photos, understanding speech, translating languages and detecting diseases in medical images. It improves as it is trained on more data.

(ix) What makes big data valuable?

Big data is valuable because its huge volume, variety and speed reveal patterns no small sample can show, for example what customers want, where diseases spread, or how traffic moves. Businesses and governments use those patterns to make better, faster decisions.

(x) What do you know about blockchain data storage?

Blockchain stores data in blocks linked in a chain, where each block carries the fingerprint (hash) of the previous block. Once a block is added it is very hard to change without changing every later block, so the record is secure, transparent and trusted without a single central owner.

3. Detailed answers.

(i) Define and differentiate between data science, data analytics, computer science, mathematics, statistics, and business knowledge.

Computer science builds computers and programs; mathematics provides the logic and formulas; statistics provides methods to collect, summarise and judge data; data analytics examines data to answer a specific question; data science combines all of these to find patterns in big data; and business (domain) knowledge tells the analyst what the results

mean for a real organisation and which decisions they support.

(ii) Explain the role of data science in different fields of life.

In healthcare data science predicts disease and plans treatment; in business it studies customers and sales; in education it tracks student progress; in transport it plans routes and traffic; in agriculture it predicts weather and crop yields; and in government it plans services from census and survey data.

(iii) Describe two business problems that could benefit from data science.

First, a shop can use past sales data to predict which products will sell next month and stock the right amounts, avoiding waste and shortages. Second, a bank can use transaction patterns to detect unusual, possibly fraudulent activity quickly and protect customers' money.

(iv) Define data storage. How can we retrieve stored data?

Data storage is keeping data on a medium such as a hard disk, SSD, database or cloud server so it can be used later. To retrieve it, a user or program sends a request or query (for example a database query or a file search), the system locates the data, and returns it for display or further processing.

(v) Define the characteristics of big data. Give an example of big data and how a business uses it to make better decisions.

Big data is described by volume (very large amounts), velocity (arriving very fast), variety (text, numbers, images, video), veracity (accuracy must be checked) and value (useful patterns). For example, an online shop analyses millions of clicks and purchases to recommend products each customer is likely to buy, which raises sales and customer satisfaction.

Unit 7: Applications of Computer Science

Exercise 7

1. Multiple choice questions. (Correct option marked with a green tick.)

(i) Which is the computer software designed to tackle complex problems and make decisions similar to those made by human experts?

(a) Data Analytics

(b) Expert System ✓

(c) Deep Learning

(d) Artificial Intelligence

(ii) Which of the following uses the medical history and current situation of a particular human being to predict future diseases?

(a) Data Analytics

(b) Expert System

(c) Deep Learning

(d) Artificial Intelligence ✓

(iii) Supervised machine learning uses _____.

(a) Labelled Dataset ✓

(b) Un-labelled Dataset

(c) Mathematics

(d) None of these

(iv) Which of the following refers to any machine that exhibits human-like characteristics such as learning and problem-solving?

(a) Computer Science

(b) Artificial Intelligence ✓

(c) Machine Learning

(d) Data Science

(v) Which of the following is a subset of machine learning that relies on artificial neural network design?

(a) Robotics

(b) Artificial Intelligence

(c) Deep Learning ✓

(d) Data Mining

(vi) Finding underlying patterns, similarities, or clusters in the data is frequently the main objective of _____.

(a) Unsupervised Learning ✓

(b) Supervised Learning

(c) Robotics

(d) Deep Learning

(vii) Which of the following is an abstraction of computing, storage, and network infrastructure?

(a) Robotics

(b) Artificial Intelligence

(c) Cloud Computing ✓

(d) Grid Computing

(viii) Amazon EC2 is an example of _____.

(a) Public Cloud ✓

(b) Private Cloud

(c) Hybrid Cloud

(d) All of these

2. Short answers.

(i) List some applications of AI.

Applications of AI include expert systems for advice, robots in factories and hospitals, speech recognition in phones, computer vision in cameras and cars, chatbots for customer service, and recommendation systems in online shops and video services.

(ii) Define robotics. Also list some uses of robotics.

Robotics is the field that designs, builds and uses robots: machines that sense, decide and act in the physical world. Robots are used in factories for assembly, in hospitals for surgery assistance, in farms, in warehouses, and in dangerous places such as mines or disaster areas.

(iii) What are the benefits and uses of speech recognition?

Speech recognition converts spoken words into text or commands, so users can dictate messages, control a phone hands-free, and use voice assistants. It helps people with disabilities, speeds up typing, and is used in call centres, cars and smart homes.

(iv) What is cloud computing and the need for it?

Cloud computing is using computing, storage and network resources over the Internet instead of owning them, so applications can be deployed quickly and scaled up or down as needed. It is needed because it saves the cost of buying hardware, allows work from anywhere, and handles sudden growth in users.

(v) List some ethical issues in some popular AI tools.

Ethical issues in AI tools include bias in training data that leads to unfair decisions, loss of privacy when personal data is collected, spread of false information by generated text and images, job loss from automation, and unclear responsibility when an AI system makes a harmful mistake.

(vi) In what ways does computer vision support task automation?

Computer vision lets machines understand images and video, so factories can inspect products automatically, cars can read road signs, phones can unlock by face, farms can spot diseased crops, and security cameras can detect unusual activity without a person watching every screen.

(vii) Describe how AI contributes to advancements in military and defence.

In defence AI helps analyse satellite and sensor data, plan logistics, detect threats, guide unmanned vehicles and drones, and train soldiers with simulations. These uses raise accuracy and speed, but they also raise serious ethical and safety questions about human control.

(viii) Differentiate between public and private cloud.

A public cloud (such as Amazon EC2) is shared over the Internet by many customers and run by a provider, so it is cheap and scalable. A private cloud is used by one organisation only, giving more control and security but at higher cost. A hybrid cloud combines both.

3. Detailed answers.

(i) Define and describe the types of machine learning.

Machine learning is a part of AI in which a computer learns patterns from data instead of being programmed for every case. In supervised learning the model is trained on a labelled dataset, for example diagnosed cases to predict disease. In unsupervised learning the model finds patterns, similarities or clusters in unlabelled data, for example grouping customers by buying habits.

(ii) What are the applications of natural language processing with suitable examples?

Natural language processing (NLP) lets computers understand human language. Applications include machine translation (Google Translate), chatbots and voice assistants (answering questions in plain language), sentiment analysis (judging opinions in reviews), text summarisation, and spell and grammar checking in word processors.

(iii) Relate how AI can be beneficial in business and education with appropriate examples.

In business AI recommends products to customers, answers queries through chatbots, predicts sales and detects fraud. In education AI gives personalised practice to each student, marks quizzes instantly, answers common questions, and helps teachers spot students who need extra help.

(iv) Explain AI and machine learning along with the problems each can handle.

AI is the broad field of making machines exhibit human-like learning, reasoning and problem-solving, handling problems such as expert advice, vision, speech and robotics. Machine learning is a part of AI that learns patterns from data, handling problems such as predicting prices, classifying emails as spam, grouping customers, and recognising images, where rules are hard to write by hand.

(v) Evaluate the social implications of the usage of Artificial Intelligence (AI) in effective decision making.

AI can make decisions faster and more consistently than people, improving healthcare, business and public services. At the same time it can entrench bias from its training data, reduce transparency when decisions cannot be explained, displace jobs, and concentrate power in whoever owns the data and systems. Responsible use therefore needs fairness checks, human oversight, privacy protection and clear accountability.

Unit 8: Impacts of Computing

Exercise 8

1. Multiple choice questions. (Correct option marked with a green tick.)

(i) Responsible hardware use involves taking precautions to ensure the longevity and safety of:

(a) Data

(b) Devices ✓

(c) Files

(d) Internet

(ii) Which of the following protects people's privacy and ensures corporations utilize data responsibly?

(a) Regulations ✓

(b) Rules

(c) Ethics

(d) Trade secrets

(iii) Keeping software updated with the latest security patches falls under:

(a) Responsible use of hardware

(b) Responsible use of online platforms

(c) Responsible use of data

(d) Responsible use of software ✓

(iv) Which of the following protects logos, sounds, words, colors, or symbols used by a company?

(a) Phishing

(b) Trademarks ✓

(c) Piracy

(d) Trade secrets

(v) Which of the following is the right of the original creator of original works of intellectual property?

(a) Copyright law ✓

(b) Computer law

(c) Phishing

(d) Innovation

(vi) Which is the invention and use of unique ideas, technologies, and techniques in the realm of computing?

(a) Invention

(b) Innovation ✓

(c) Automation

(d) Online Learning

(vii) Which is an attack that involves the unlawful copying, distribution and use of software?

(a) Soft lifting

(b) Cyberstalking

(c) Software piracy ✓

(d) Hard disk overloading

(viii) Which is the use of the Internet or other online resources to harass an individual/organization?

(a) Piracy

(b) Cyberstalking ✓

(c) Money Laundering

(d) Cyber extorsion

(ix) Which is the use of internet platforms and websites to interact and communicate with people?

(a) Social Networking ✓

(b) Innovation

(c) Automation

(d) None of these

(x) Which of the following can help businesses become more efficient and save time, money, and resources?

(a) Computer Laws

(b) Security

(c) Automation ✓

(d) Online Scams

2. Short answers.

(i) Identify the responsible use of computing resources.

Responsible use means using hardware carefully, keeping software updated, protecting passwords and personal data, respecting the law and other users, avoiding piracy, and switching off or recycling devices properly instead of wasting energy and equipment.

(ii) Define intellectual property rights.

Intellectual property rights are legal rights that protect creations of the mind, such as books, music, software, inventions and brand names. They give the creator control over copying and use, and include copyright, patents, trademarks and trade secrets.

(iii) List the different drawbacks of computing innovation.

Drawbacks of computing innovation include job loss from automation, privacy loss from data collection, cybercrime and scams, addiction and health problems from overuse,

electronic waste, and the spread of false information online.

(iv) Identify malicious internet scams.

Malicious internet scams include phishing (fake messages that steal passwords or money), software piracy, hacking, spamming (bulk unwanted messages), cyberstalking (online harassment) and online fraud such as fake shops or prize messages.

(v) Differentiate between patent and trade secret.

A patent is a public legal right granted for a new invention, giving the inventor exclusive use for a fixed period in exchange for publishing how it works. A trade secret is confidential business information, such as a recipe, that is kept secret and protected only as long as it stays secret.

(vi) What is the importance of computer laws?

Computer laws protect users, data and creators: they punish hacking, piracy and online fraud, protect privacy and intellectual property, and give businesses and citizens confidence to use computers and the Internet safely.

(vii) What is online security and privacy?

Online security means protecting computers, accounts and data from attacks, theft and damage, using strong passwords, updates and antivirus software. Online privacy means controlling what personal information is collected and shared, and by whom.

(viii) Identify the key benefits of mobile technology.

Mobile technology lets people communicate, learn, work, bank and shop from anywhere at any time. It gives instant access to information, maps and services, supports remote work and education, and connects people in remote areas.

3. Detailed answers.

(i) Discuss how ethics are guidelines for responsible use of computing systems.

Ethics are moral guidelines for right and wrong use of computers. They require honesty (no piracy or fraud), respect for privacy (no snooping or sharing others' data), responsibility (keep systems secure and updated), fairness (no discrimination or harassment) and care for shared resources. Following ethics keeps users, organisations and society safe.

(ii) Explain the computer-related laws with the help of appropriate examples.

Copyright law protects original works, so copying a paid software or a film without permission is illegal. Patent law protects new inventions, such as a new device design. Trademark law protects brand logos and names, so a fake shop cannot use a famous brand's logo. Trade secret law protects confidential information, such as a company's secret formula, from theft.

(iii) Define social networking. Discuss the positive and negative impacts of social networking.

Social networking is using internet platforms and websites to interact and communicate with people, for example sharing updates, photos and messages. Positives: staying in

touch, learning, business promotion and community support. Negatives: privacy loss, cyberbullying, addiction, false information and wasted time.

(iv) Examine the different ways in which computing innovations have had an impact on society.

Computing innovations changed how we work (remote work and automation), learn (online classes), communicate (instant messaging and video calls), shop and bank (online services), receive healthcare (telemedicine and digital records) and entertain ourselves (streaming and games). They raised productivity but also created privacy, security and inequality challenges.

(v) What is computing innovation? Discuss the benefits of computing innovation in detail.

Computing innovation is the invention and use of new ideas, technologies and techniques in computing, such as AI, cloud computing and mobile apps. Its benefits are faster work, lower costs, better communication, new jobs and businesses, improved healthcare and education, and solutions to problems that were previously hard or impossible to solve.

Unit 9: Entrepreneurship in Digital Age

Exercise 9

1. Multiple choice questions. (Correct option marked with a green tick.)

(i) The process of setting up a new business is known as:

(a) New business

(b) Innovation

(c) Intrapreneurship

(d) Entrepreneurship ✓

(ii) An individual who creates a new business, bearing most of the risks and enjoying the rewards is called:

(a) Shopkeeper

(b) Entrepreneur ✓

(c) Student

(d) Innovators

(iii) Which of the following factors provides growth in business?

(a) Environment

(b) Technology ✓

(c) Business idea

(d) Workers

(iv) Which of the following is a method for solving complex problems that takes into account the user's needs and emotions?

(a) Deep thinking

(b) Design thinking ✓

(c) Online Business

(d) Over thinking

(v) In which stage of design thinking do you identify the best possible solution?

(a) Ideate ✓

(b) Test

(c) Prototype

(d) Define

(vi) A business plan is like a written summary of:

(a) business aims and objectives

(b) business strategies

(c) operational plans to achieve its goals

(d) All of these ✓

(vii) Entrepreneurs utilize a range of _____ to promote their businesses.

(a) Traditional marketing platforms

(b) Digital marketing platforms ✓

(c) Telemarketing

(d) None of these

(viii) Which of the following meets the needs of the present, without compromising the ability of future generations to meet their own needs?

(a) Innovation

(b) Digital marketing

(c) Sustainable development ✓

(d) Design thinking

2. Short answers.

(i) Define Entrepreneurship.

Entrepreneurship is the process of setting up a new business: spotting an opportunity, taking the risk, organising resources and creating a product or service to earn a reward. The person who does this is called an entrepreneur.

(ii) List any three roles of an entrepreneur.

Three roles of an entrepreneur are: (1) taking risks and investing money and effort, (2) organising resources such as workers, materials and finance, and (3) innovating by creating new products, services or better ways of working.

(iii) Describe the entrepreneurial process briefly.

The entrepreneurial process begins with finding an opportunity and an idea, then planning (including a business plan), then gathering resources and starting the business, then running and growing it, and finally reviewing results and improving or expanding.

(iv) Define the design thinking process.

Design thinking is a human-centred method for solving complex problems that takes the user's needs and emotions into account. It works in five phases: Empathise, Define, Ideate, Prototype and Test, repeating the cycle until the solution truly fits the user.

(v) List down two benefits of the design thinking process.

Two benefits are: (1) solutions fit real user needs, because the process starts by understanding users, and (2) ideas are tested early with prototypes, so mistakes are found cheaply before a full product is built.

(vi) Define business plan.

A business plan is a written summary of a business: its aims and objectives, its strategies, its products and market, and its operational and financial plans to achieve its goals. It guides the entrepreneur and helps win finance or partners.

(vii) List some digital marketing platforms used by entrepreneurs.

Digital marketing platforms used by entrepreneurs include Facebook, Instagram, YouTube, TikTok, WhatsApp Business, Google Ads and their own websites or online shops, where they

advertise, take orders and talk to customers.

(viii) What is meant by sustainable development? Give some examples.

Sustainable development means meeting the needs of the present without compromising the ability of future generations to meet their own needs. Examples: using solar energy instead of fossil fuels, recycling and reducing waste, planting trees, and saving water.

3. Detailed answers.

(i) How to define and articulate an entrepreneurial solution to address a local or global problem?

First identify a real local or global problem and the people it affects, for example plastic waste in a town. Then define the solution clearly: what product or service will solve it, who will use it, and why it is better than existing options. Finally articulate it in a short pitch and business plan stating the problem, solution, market, cost and benefit.

(ii) Why do we use the design thinking method in our daily life? Explain.

We use design thinking in daily life because many problems are complex and involve other people's needs, not just our own idea. By empathising, defining the real problem, generating many ideas, prototyping and testing, we avoid building the wrong solution and reach one that people actually want and can use.

(iii) What are the different phases of the design thinking process? Explain.

The five phases are: (1) Empathise — understand users' needs and feelings; (2) Define — state the problem clearly in human-centred words; (3) Ideate — generate many possible solutions; (4) Prototype — build a simple, cheap version of the best idea; (5) Test — try it with users, learn, and improve or repeat earlier phases.

(iv) How to collect and synthesize information to create a business plan?

Collect information by market research: surveys and interviews with customers, studying competitors, checking costs, prices and suppliers, and noting legal requirements. Then synthesise it: keep only what matters, organise it into aims, market, product, operations and finance sections, and write clear targets and steps the business will follow.

(v) How can we achieve the sustainable development goals?

Sustainable development goals can be achieved by using clean energy, reducing waste and recycling, protecting water and forests, providing quality education and healthcare for all, promoting decent work and equality, and by governments, businesses and individuals each taking responsibility for their own footprint.